

CSA0501

DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENT

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EXPERIMENT – 15

Aim

To learn and implement DCL commands GRANT and REVOKE in MySQL.

Procedure

1. Open MySQL Workbench.
2. Connect to your MySQL server.
3. Select BikeShowroomDB.
4. Create a separate MySQL user for the experiment.
5. Grant privileges to the user.
6. Verify the granted privileges.
7. Revoke the privileges.
8. Verify the revoked privileges.

The manual defines GRANT as providing privileges and REVOKE as taking back privileges.

QUESTIONS & OUTPUTS :

1. Create a user for performing DCL operations.
2. Grant all privileges on your BikeShowroomDB database to the user.
3. Insert another bike and create another savepoint

1. Create a user for performing DCL operations.

The screenshot shows a MySQL IDE with a SQL editor containing the following commands:

```
1 • CREATE USER 'bikeuser'@'localhost'  
2 IDENTIFIED BY 'bike123';  
3 • SELECT User, Host  
4 FROM mysql.user  
5 WHERE User = 'bikeuser';
```

Below the editor, the 'Result Grid' shows the output of the second query:

User	Host
bikeuser	localhost

The 'Output' pane shows the execution log:

#	Time	Action	Message
1	13:32:10	CREATE USER bikeuser@localhost IDENTIFIED BY bike123	0 row(s) affected
2	13:32:10	SELECT User, Host FROM mysql.user WHERE User = 'bikeuser' LIMIT 0, 1000	1 row(s) returned

2. Grant all privileges on your BikeShowroomDB database to the user.

The screenshot shows a MySQL IDE with a SQL editor containing the following commands:

```
1 • GRANT ALL PRIVILEGES  
2 ON BikeShowroomDB.*  
3 TO 'bikeuser'@'localhost';  
4 • SHOW GRANTS  
5 FOR 'bikeuser'@'localhost';
```

Below the editor, the 'Result Grid' shows the output of the third query:

Grants for bikeuser@localhost	
GRANT USAGE ON *.* TO 'bikeuser'@'localhost'	
GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...	

The 'Output' pane shows the execution log:

#	Time	Action	Message
1	13:32:41	GRANT ALL PRIVILEGES ON BikeShowroomDB.* TO bikeuser@localhost	0 row(s) affected
2	13:32:41	SHOW GRANTS FOR bikeuser@localhost	2 row(s) returned

3. Give the user permission to view the bike_data table.

```
1 • GRANT SELECT
2 ON BikeShowroomDB.bike_data
3 TO 'bikeuser'@'localhost';
4 • SHOW GRANTS
5 FOR 'bikeuser'@'localhost';
```

Result Grid			
Grants for bikeuser@localhost			
▶ GRANT USAGE ON *.* TO 'bikeuser'@'localho...			
GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...			
GRANT SELECT ON 'bikeshowroomdb'. 'bike_d...			

Result 3			
Read Only			
Output			
Action Output			
#	Time	Action	Message
✓ 1	13:34:41	GRANT SELECT ON BikeShowroomDB bike_data TO 'bikeuser'@'localhost'	0 row(s) affected
✓ 2	13:34:41	SHOW GRANTS FOR 'bikeuser'@'localhost'	3 row(s) returned

4. Give the user permission to insert records into bike_data.

```
1 • GRANT INSERT
2 ON BikeShowroomDB.bike_data
3 TO 'bikeuser'@'localhost';
4 • SHOW GRANTS
5 FOR 'bikeuser'@'localhost';
```

Result Grid			
Grants for bikeuser@localhost			
▶ GRANT USAGE ON *.* TO 'bikeuser'@'localho...			
GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...			
GRANT SELECT, INSERT ON 'bikeshowroomdb'...			

Result 4			
Read Only			
Output			
Action Output			
#	Time	Action	Message
✓ 1	13:35:14	GRANT INSERT ON BikeShowroomDB bike_data TO 'bikeuser'@'localhost'	0 row(s) affected
✓ 2	13:35:14	SHOW GRANTS FOR 'bikeuser'@'localhost'	3 row(s) returned

5. Give the user permission to update records in bike_data.

```
1 • GRANT UPDATE
2   ON BikeShowroomDB.bike_data
3   TO 'bikeuser'@'localhost';
4 • SHOW GRANTS
5   FOR 'bikeuser'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Grants for bikeuser@localhost

- GRANT USAGE ON *.* TO 'bikeuser'@'localho...
- GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...
- GRANT SELECT, INSERT, UPDATE ON 'bikesho...

Result 5 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 1	13:35:52	GRANT UPDATE ON BikeShowroomDB.bike_data TO bikeuser@'localhost'	0 row(s) affected
✓ 2	13:35:52	SHOW GRANTS FOR 'bikeuser'@'localhost'	3 row(s) returned

6. Give the user permission to delete records from bike_data.

```
1 • GRANT DELETE
2   ON BikeShowroomDB.bike_data
3   TO 'bikeuser'@'localhost';
4 • SHOW GRANTS
5   FOR 'bikeuser'@'localhost';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Grants for bikeuser@localhost

- GRANT USAGE ON *.* TO 'bikeuser'@'localho...
- GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...
- GRANT SELECT, INSERT, UPDATE, DELETE ON ...

Result 6 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 1	13:36:22	GRANT DELETE ON BikeShowroomDB.bike_data TO bikeuser@'localhost'	0 row(s) affected
✓ 2	13:36:22	SHOW GRANTS FOR 'bikeuser'@'localhost'	3 row(s) returned

7. Now take back the SELECT permission from the user.

The screenshot displays the SQL Developer interface. The top pane shows the following SQL commands:

```
1 • REVOKE SELECT
2   ON BikeShowroomDB.bike_data
3   FROM 'bikeuser'@'localhost';
4 • SHOW GRANTS
5   FOR 'bikeuser'@'localhost';
```

The bottom pane shows the results of the execution. The 'Result Grid' tab is active, displaying the grants for the user 'bikeuser'@'localhost':

Grants for bikeuser@localhost
GRANT USAGE ON *.* TO 'bikeuser'@'localho...
GRANT ALL PRIVILEGES ON 'bikeshowroomdb'...
GRANT INSERT, UPDATE, DELETE ON 'bikesho...

Below the Result Grid, the 'Action Output' tab is active, showing the execution log:

#	Time	Action	Message
✓ 1	13:37:00	REVOKE SELECT ON BikeShowroomDB bike_data FROM bikeuser@localhost	0 row(s) affected
✓ 2	13:37:00	SHOW GRANTS FOR 'bikeuser'@'localhost'	3 row(s) returned

RESULT

Thus, the DCL commands GRANT and REVOKE were successfully executed to provide and remove privileges for the bikeuser on the BikeShowroomDB database